

MATH 2211, SPRING 2026: FINAL STUDY GUIDE

This exam will consist of 8 sections, the first of which will consist of true or false questions. For the latter, you will be expected to not only give the right response, but also a correct justification for that response, in one or two sentences.

You will have the full 3 hours to work on the exam. Make sure to at least *attempt* every problem in a substantial way.

You will be allowed to bring two pages of notes. By this, I mean both sides of a single sheet of *Letter-sized* paper, or one side each of two sheets of *Letter-sized* paper.

The exam will be cumulative and will cover the following topics:

- (1) Proofs by induction (usual and strong): This involves clearly setting up the base case, the inductive hypothesis, and then the inductive step, with the use of the inductive hypothesis also made in comprehensible fashion. See Lecture 2 and Homework 1 for examples.
- (2) Modular arithmetic: Lecture 4
- (3) Fields and their properties:
 - (a) Definition: Lecture 4 and Lecture 5.
 - (b) Examples of various fields: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}/p\mathbb{Z}$ (p prime), $\mathbb{C}$: Lecture 5.
 - (c) Arithmetic in the complex numbers: Modulus/absolute value, Euler's notation, multiplication using Euler's notation. See Lecture 5 and Lecture 6.
 - (d) Basic properties of fields: Problem 3 of Homework 2.
- (4) Linear systems of equations over fields using row operations and their applications: This appears everywhere, including in Lecture 3, Lecture 6 (finding polynomials with certain properties), Lecture 10 (understanding if one vector is in the span of some other vectors).
- (5) Vector spaces and their properties:
 - (a) Definition: Lecture 6
 - (b) Examples: Lecture 7. Note in particular, the vector spaces F^n , $\text{Poly}(F)$, $\text{Poly}_{\leq m}(F)$, $M_{n \times m}(F)$, the last of which was introduced in Lecture 8.
 - (c) Basic properties: Lecture 7.
 - (d) Subspaces: Lecture 7
- (6) Span of subsets $X \subset V$: Lecture 8
 - (a) Definition.
 - (b) Linear combinations.
 - (c) How to decide if a vector is in the span of another set of vectors: see also Example in Lecture 9.
- (7) Linear independence: Definition and properties. See Lecture 9.
- (8) Bases: Lecture 10.
 - (a) Definition.
 - (b) The standard basis for F^n .
 - (c) Examples.
- (9) The Steinitz Exchange Lemma and its consequences:
 - (a) Statement of the lemma: Lecture 11
 - (b) Corollary 1 from Lecture 11: This explains what it means to be finite dimensional and also the well-definedness of the notion of dimension. You should understand the proof and how the Exchange Lemma is used.
 - (c) Proof of the lemma: Lecture 12
- (10) Finite dimensional spaces and dimension: Lecture 12.
 - (a) Definitions
 - (b) Examples
 - (c) Make note of Observations 1, 2, 3 here, which establish useful properties of bases that we will return to often.

- (11) Functions and their properties:
 - (a) Definition: Lecture 13
 - (b) One-to-one or injective; onto or surjective: Lecture 13
 - (c) Composition: Lecture 13
 - (d) Bijectivity and existence of inverse function: Lecture 13.
- (12) Linear transformations:
 - (a) Definition and basic properties: Lecture 14
 - (b) Invertibility: Lecture 14
 - (c) Linear transformations with F^n as domain: Lecture 14
 - (d) Linear transformations from F^n to F^m are given by $n \times m$ -matrices: Lecture 15
 - (e) Invertible linear transformations or isomorphisms: Lecture 16

You should be very familiar with especially part (d)
- (13) Kernel and image:
 - (a) Definitions: Lecture 18.
 - (b) Finding bases for kernel and image using row reduction; reduced row echelon form: Lecture 18
 - (c) Rank-nullity: Lecture 18
 - (d) Injectivity and vanishing of kernel: Fact in Lecture 19
 - (e) Criteria for a linear transformation between vector spaces of the same dimension to be invertible: Lecture 19
- (14) Matrix multiplication:
 - (a) Definition of Av : Lecture 16
 - (b) Definition of AB when A is $n \times p$ and B is $p \times m$ and relationship with composition of linear transformations: Homework 5, Problem 7
 - (c) See more detailed description of matrix products in terms of dot products and transposed row vectors in Lecture 20
 - (d) Invertible matrices and matrix inverses; deciding invertibility and finding the inverse using RREF: Homework 5, Problem 1
- (15) Bases, linear transformations and matrices:
 - (a) Connection between bases for V and isomorphisms $F^n \xrightarrow{\cong} V$: Lecture 16 and Homework 5, Problem 8
 - (b) Going from linear transformations to matrices via bases: $T \mapsto [T]_{\mathcal{B}_1, \mathcal{B}_2}$. See Lecture 16. The main case we will be concerned with is when $V = W$ and $\mathcal{B} = \mathcal{B}_1 = \mathcal{B}_2$, when we just write $[T]_{\mathcal{B}}$.
 - (c) Change of basis: Going from $[T]_{\mathcal{B}}$ to $[T]_{\mathcal{B}'}$ via *conjugation* by a change of basis matrix. Initial discussion is in Lecture 17, and more detailed discussion is in Lecture 23
- (16) The determinant and its properties:
 - (a) The 2×2 determinant and its connection with invertibility: Proposition in Lecture 21
 - (b) Key properties of 2×2 determinant: Lecture 23
 - (c) Uniqueness of the general $n \times n$ determinant: Proposition in Lecture 24
 - (d) Behavior of determinant under row operations: Lecture 24
 - (e) Computing the determinant using row operations: Lecture 24
 - (f) Compatibility of determinant with products: Lecture 24 and Homework 8, Problem 2
 - (g) Elementary matrices and transpose invariance of determinant: Lecture 25
 - (h) Defining the $n \times n$ determinant via expansion by minors: See Lecture 26 for expansion along first column, and Problem 6 on Homework 9 for expansion along other columns and rows.
 - (i) Cramer's rule for the inverse: Problem 7 on Homework 9.
- (17) Diagonalizability:
 - (a) Diagonal matrices and their products: Lecture 20
 - (b) Definition of diagonalizable linear transformation: Lecture 20
 - (c) Characteristic polynomial and eigenvalues for 2×2 matrices: Lecture 22
 - (d) Deciding if a matrix is diagonalizable using its eigenvalues: Lecture 22
 - (e) If matrix is diagonalizable, using appropriate basis to compute its powers: Lecture 23 and Homework 7, Problem 4.
 - (f) Characteristic polynomial for $n \times n$ matrices: Definition 3 on Lecture 27.

- (g) Relationship between characteristic polynomial and eigenvalues: Observation 2 on Lecture 27.
- (h) Characteristic polynomial and eigenvalues for upper triangular matrices: Observation 5 on Lecture 27.
- (i) Conjugation invariance of the characteristic polynomial: Problem 1 of Homework 9.
- (j) Eigenbases and diagonalizability: Lecture 28
- (k) Linear independence of eigenvectors with distinct eigenvalues: Proposition 1 on Lecture 28 and Lecture 29.
- (l) The minimal polynomial: Definition 2 on Lecture 30
- (m) The minimal polynomial divides all polynomials that kill A : Remark 2 on Lecture 30
- (n) Conjugation invariance of the minimal polynomial: Observation 3 on Lecture 31
- (o) Characterizing diagonalizability in terms of the minimal polynomial having distinct factors: Proposition on Lecture 31
- (p) Algebraic and geometric multiplicity: Definitions 1,2,3 on Lecture 31.
- (q) Characterization of eigenvalues in terms of non-zero algebraic or geometric multiplicity: Remark on Lecture 31.
- (r) Generalized eigenspaces and generalized eigenvectors: Lecture 37 and also Homework 10.
- (18) Orthogonal projections and least squares:
 - (a) Orthogonal complements: Definition on Lecture 33.
 - (b) Relation between the image of A and the kernel of A^T via orthogonal complements: Observation on Lecture 33.
 - (c) Least squares solution: Definition on Lecture 33.
 - (d) Orthonormal basis for a subspace of $\mathbb{R}^n$: Definition on Lecture 34
 - (e) The Gram-Schmidt theorem and algorithm: Lecture 34.
 - (f) Orthogonal matrices: Definition on Lecture 34
 - (g) Understanding Gram-Schmidt algorithm in terms of orthogonal projections: Lecture 35
 - (h) The QR decomposition and using Gram-Schmidt to obtain it: Proposition on Lecture 36
 - (i) Least squares using the QR decomposition: Observation 2 on Lecture 36